

Modern Cryptographic Solutions

Objectives:

At the end of this episode, I will be able to:

1. Describe the difference between symmetric and asymmetric algorithms
2. Describe hashing and HMAC
3. Describe the overall purpose and approach to technologies like IPsec and SSL/TLS

Additional resources used during the episode can be obtained using the download link on the overview episode.

Here are some key points you should remember from this episode:

- Symmetric encryption algorithms use the same key for encryption and decryption
- Asymmetric encryption algorithms use a key pair for the encryption and decryption processes
- Symmetric are excellent algorithms for large amounts of bulk data encryption
- Asymmetric are excellent for tasks like initial secure authenticated sessions; then symmetric algorithms can safely perform the bulk data encryption

External Resources:

During this episode, you can reference the following external resources for supplementary tools and information:

Here is a great read on the next generation security protocols. This is the Suite B information we discussed during the episode:

https://www.cisco.com/c/dam/en_us/solutions/industries/docs/gov/CIS-44830_Suite-B_Whitepaper_1c-hi_res.pdf